

Seungmin Lee

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Research Areas

- Building-grid resilience, including weather-driven outage forecasting and restoration dynamics
- Machine learning and time-series analytics for power systems and building energy applications
- Building electrification, heat-pump load identification, and distribution-system planning

Education

Ph.D. Student in Design (Building Energy System) North Carolina State University, Raleigh, NC, USA Advisor: Professor Soolyeon Cho	08/2023 – Present
M.S. Student in Electrical and Computer Engineering North Carolina State University, Raleigh, NC, USA	08/2025 – Present
M.E. in Architectural Engineering Inha University, Incheon, Korea Advisor: Professor Euijong Kim GPA 3.89/4.0 Thesis [T1]	03/2013 – 08/2019
B.S. in Architecture Konyang University, Nonsan, Korea GPA 3.53/4.0	03/2009 – 02/2013

Publications & Thesis

- P5)** (Under review) **Seungmin Lee**, Juwan Ha, Wenyuan Tang, Soolyeon Cho. Forecasting Rare, Large-Scale Weather-Induced Power Outages Using Temporal Fusion Transformers: A Multi-County Study for Building Energy Resilience. *Science and Technology for the Built Environment*.
- P4)** (Preparing for submission) **Seungmin Lee**, Juwan Ha, Wenyuan Tang, Soolyeon Cho. From Outage Duration to Recovery Pathways: A Review of Integrated Building-Grid Restoration Dynamics. Expected 2026.
- P3)** **Seungmin Lee**, Seunghoon Park, Yongsung Jang, and Euijong Kim. (2021). Proposition of design capacity of borehole heat exchangers for use in the schematic-design stage. *Energies*, 14(4): 822.
- P2)** **Seungmin Lee**, and Euijong Kim. (2019). Analysis of GHG reduction in apartment buildings by 2030 according to various prediction scenarios. *Journal of Korea Institute of Architectural Sustainable Environment and Building Systems*, 13(6): 515–526.
- P1)** **Seungmin Lee**, Jinho Kim, Gwangsu Shin, and Euijong Kim. (2018). Market acceptability of the ZEB certification system for public buildings according to the 2025 roadmap. *Journal of Korea Institute of Architectural Sustainable Environment and Building Systems*, 12(6): 557–566.
- T1)** (Master's Thesis) **Seungmin Lee**. (2018). Evaluation of GHG reduction in residential buildings according to proposed scenarios, Inha University.

Conference Presentations

- C13)** **Seungmin Lee**, Juwan Ha, Yujin Kim, Soolyeon Cho. (2026). Forecasting High-Impact, Low-Probability Power Outages with Temporal Fusion Transformers: A Case Study of Wake County. *2026 ASHRAE Winter Conference*, Las Vegas, NV, USA.
- C12)** Juwan Ha, **Seungmin Lee**, Yujin Kim, Soolyeon Cho. (2026). Model Predictive Coordinated Control for Grid-Interactive Residential Building with Energy Recovery Ventilator, PV, and Battery Energy Storage System. *2026 ASHRAE Winter Conference*, Las Vegas, NV, USA.
- C11)** Yujin Kim, Juwan Ha, **Seungmin Lee**, Soolyeon Cho. (2026). Ontology-Based Semantic Modeling of HVAC Systems in the Office Building: A Foundation for Scalable Control and Digital Twin Integration. *2026 ASHRAE Winter Conference*, Las Vegas, NV, USA.
- C10)** Juwan Ha, **Seungmin Lee**, Yujin Kim, Soolyeon Cho. (2025). Improving Energy and Data Efficiency in Factory Buildings: A Physics-Informed Neural Networks-Based Model Predictive Control (PINNs-Based MPC) Approach. *2025 ASHRAE Annual Conference*, Phoenix, AZ, USA.

- C9) **Seungmin Lee**, Juwan Ha, Soolyeon Cho. (2024). A review of federated learning applications in building energy management. *US-Korea Conference on Science, Technology, and Entrepreneurship (UKC)*, San Francisco, CA, USA.
- C8) Juwan Ha, **Seungmin Lee**, Soolyeon Cho. (2024). Development of heat pump prediction model based on limited experimental data in factory building. *UKC 2024*, San Francisco, CA, USA. (Poster)
- C7) **Seungmin Lee**, Juwan Ha, Jonghoon Kim, Soolyeon Cho. (2023). An Empirical Analysis of Korean Household Appliance Use Patterns: using a national Time Use survey dataset. *UKC 2023*, Dallas, TX, USA.
- C6) Juwan Ha, Byeongmo Seo, **Seungmin Lee**, Soolyeon Cho. (2023). Fast Load Prediction Model of Chiller Using Bayesian Optimization. *UKC 2023*, Dallas, TX, USA.
- C5) **Seungmin Lee**, Seunghoon Park, and Euijong Kim. (2020). A study on the calculation of heat transfer rate per borehole according to the bore field configuration. *Journal of Korea Institute of Architectural Sustainable Environment and Building Systems*, Chungju, Korea.
- C4) **Seungmin Lee**, and Euijong Kim. (2019). Analysis of GHG reduction potential for new apartment houses. *Journal of the Korean Solar Energy Society*, Daegu, Korea.
- C3) **Seungmin Lee**, and Euijong Kim. (2018). Relationship between the existing mandatory regulations and the ZEB certification roadmap. *Journal of Korea Institute of Architectural Sustainable Environment and Building Systems*, Incheon, Korea.
- C2) **Seungmin Lee**, Seunghoon Park, and Euijong Kim. (2018). Attainability analysis of ZEB certification system obligation roadmap. *Journal of the Korean Solar Energy Society*, Seoul, Korea.
- C1) **Seungmin Lee**, Hyangin Jang, Cheolyong Jang, and Seungjik Suh. (2013). A Study on the Applicability of Cool Roof in the Domestic Climate According to Energy Consumption Analysis. *The Society of Air-Conditioning and Refrigerating Engineers of Korea*, Seoul, Korea.

Research Projects

Development of EMS with Optimal Control Algorithm for Energy Efficiency Improvement in Commercial Building Using AI and Twin Technology

Korea Energy Agency, funded by the Ministry of Land, Infrastructure and Transport

- Developed an AI-based optimal control model utilizing building and environmental data to automate EMS in factory buildings, targeting energy savings, operational efficiency, and anomaly detection

Optimal Design Analysis of Geothermal Heat Pump Systems for High-Density City Zones Based on Detailed Modeling and Simulation

National Research Foundation (NRF) of Korea (No. 2016R1C1B2011097)

- Proposed an optimal design method for large-scale urban buildings through detailed TRNSYS simulations (thermal storage, design conditions, groundwater effects)
- Related outputs: [P3], [C5]

Appropriate Targets for Mandatory Zero-Energy Buildings and Plans to Link the Emission Trading System

Korea Energy Agency, funded by the Ministry of Land, Infrastructure and Transport

- Related outputs: [P1], [P2], [C2], [C3], [C4]

Work Experience

Graduate Intern

05/2026 – Present

Grid Automation and Controls Group, National Laboratory of the Rockies (NLR), Golden, CO, USA

- Identify heat-pump loads at the distribution feeder level using data-driven methods
- Develop load profiles for integration into transmission and distribution system models

Graduate Research Assistant

08/2023 – Present

North Carolina State University, Raleigh, NC, USA

- Study building-grid coupling by integrating building and distribution system models
- Develop and extend co-simulation approaches for building-grid interaction analysis
- Analyze building load dynamics (e.g., thermal behavior, CLPU) and impacts on grid operations
- Develop data-driven modeling approaches for grid-interactive building systems

Postgraduate Intern

09/2022 – 05/2023

Korea Institute of Energy Research (KIER), Daejeon, Korea

- Participated in the low-carbon energy-efficient remodeling (retrofitting) research project
- Evaluated changes in typical occupancy schedules for residential buildings in Korea through cluster analysis using machine learning algorithms

Postgraduate Researcher

03/2020 – 12/2020

Inha University, Incheon, Korea (Advisor: Dr. Euijong Kim)

- Participated in the NRF geothermal heat pump system design project; led the study published as [P3] and presented as [C5]

Graduate Teaching Assistant

09/2018 – 08/2019

Inha University, Incheon, Korea (Advisor: Dr. Euijong Kim)

- Devised lectures and class activities on the analysis of determinate and indeterminate structures; created and graded course assessments

Researcher → Senior Researcher (promoted)

03/2015 – 02/2018

Korea Research Institute of Eco-Environmental Architecture (KRIEA), Seoul, Korea

- Led energy performance assessments of 62 buildings of varying use (offices, schools, training facilities, stations)
- Advised property owners on energy conservation practices and renewable energy system integration
- Analyzed HVAC systems combining central (AHU, absorption chiller-heater, boiler, district heating/cooling) and individual (FCU, heat pump, A/C) systems

Graduate Researcher

03/2013 – 02/2015

Korea Institute of Energy Research (KIER), Daejeon, Korea

- Assessed energy performance of 17 residential buildings across heating systems (individual/central/district) and heat sources (oil, gas, electricity)
- Appraised insulation and airtightness performance through on-site inspection

Grants & Scholarships

Fellowships

- Mansour Doctoral Fellowship, NC State University (2023–2024) — \$10,000; one of 25 recipients among international Ph.D. candidates
- University Graduate Fellowship, NC State University (2023–2024) — \$4,000

Professional Development & Travel Grants

- ASHRAE Graduate Grant-In-Aid (2026–2027), ASHRAE (May 2026) — up to \$10,000 in support of graduate research in building energy and HVAC systems
- Ph.D. Travel Scholarship, NCSU — ASHRAE Winter Conference (2024, 2025; \$1,000 each)
- IIBEC Travel Fellowship, International Convention and Trade Show (2024) — \$2,500
- UKC Travel Support, Korean-American Scientists and Engineers Association (2023, 2024)

Earlier Scholarships

- Industry–University Corporate Scholarship (KIER & Inha Univ., 2013–2014); TA Scholarships (Inha Univ., 2018–2019); merit-based scholarships (Konyang Univ., 2009–2012)

Affiliations & Service

ASHRAE (American Society of Heating, Refrigerating and Air-Conditioning Engineers)

- Student Member (2023 – Present); President, NCSU Student Branch (08/2025 – 05/2026); Vice President (09/2024 – 07/2025)

KSEA (Korean-American Scientists and Engineers Association) — Student Member (2023 – Present)

IIBEC (International Institute of Building Enclosure Consultants) — Student Member (2023 – Present)

PhDSA (Ph.D. Student Association), College of Design, NCSU — Treasurer (09/2024 – 07/2025)

Technical Skills

Software: EnergyPlus, OpenDSS, TRNSYS, OpenStudio, Rhino, QGIS, Solar PV Design

Programming: Python (PyTorch, ML & NN models, PySpark), SQL, MATLAB